NOT MEASUREMENT SENSITIVE

MIL-P-85856/18
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS
COBALT-CHROMIUM-NICKEL-TUNGSTEN,
TYPE I, CLASS 1, COMPOSITION S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Cr	Ni	Fe	W	C	Mn	Si	Co	P	S
24.5 26.5	9.5 11.5	2.0 max.	7.0 8.0	0.45 0.55	1.0 max.	1.0 max.	Bal	0.04 max.	0.04 max.

(1) Powder shall be reduced by heating in a vacuum or dry hydrogen atmosphere for 2 to 4 hours at 1750° to 1850°F (955° to 1010°C). This requirement shall be waived if powder is produced by atomizing in an inert atmosphere.

Particle Size/Distribution

-270 Mesh (-53 µm)	100.0 min.	
+325 Mesh (+45 µm)	1.0 max.	
-20 µm	65.0 max.	35 min.
-5 µm	5.0 max.	

Sprayed Coating Properties

Bond Strength	32.0 MPa (4700 psi) min.
Hardness	Rockwell 15N, 80 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 10% max.

AMSC N/A
3439

FSC

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Deposit Efficiency and Deposition Rate

55% min for A/H_2 @ 6.5 lb/hr.

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Pratt & Whitney Aircraft	PWA 1316
Rolls-Royce Ltd.	MSRR 9507/23

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/18)

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